

## Scalability & Future Plan

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|----------------------|---------------------------------------------------------------------------------|
| <b>Date</b>          | 9 JULY 2026                                                                     |
| <b>Team ID</b>       |                                                                                 |
| <b>Project Name</b>  | Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 1 Mark                                                                          |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                           | Impact                                                                | Priority to Address (High / Medium / Low) |
|------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------|
| 1    | The model uses a static training dataset and requires manual retraining for updates. | Prediction accuracy may decrease as newer HDI data becomes available. | High                                      |
| 2    | The application is designed for a limited number of concurrent users.                | Performance may slow under heavy user traffic.                        | Medium                                    |
| 3    | Advanced analytical dashboards and visual reports are not available.                 | Users cannot perform detailed trend analysis or comparisons.          | Medium                                    |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                          | Proposed Upgrade / Solution                                                                       |
|------|--------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports basic public access with limited concurrent users.            | Deploy on scalable cloud infrastructure with automatic scaling and load balancing.                |
| 2    | Data Storage       | Uses local files and prediction history storage.                       | Integrate a cloud database such as PostgreSQL or MySQL for reliable and scalable data management. |
| 3    | Performance        | Prediction is performed using a locally loaded machine learning model. | Optimize the model, implement caching, and improve response time for large workloads.             |

|   |          |                                        |                                                                                      |
|---|----------|----------------------------------------|--------------------------------------------------------------------------------------|
| 4 | Security | Basic input validation is implemented. | Add user authentication, HTTPS, secure API endpoints, and role-based access control. |
|---|----------|----------------------------------------|--------------------------------------------------------------------------------------|

Future Roadmap:

| Phase   | Planned Feature / Enhancement                                                    | Target Timeline | Expected Impact                                    |
|---------|----------------------------------------------------------------------------------|-----------------|----------------------------------------------------|
| Phase 1 | Integrate the latest UNDP HDI dataset and improve prediction accuracy.           | 1–2 Months      | More accurate and up-to-date predictions.          |
| Phase 2 | Add interactive dashboards, charts, and country comparison features.             | 2–3 Months      | Better visualization and analysis of HDI trends.   |
| Phase 3 | Deploy on a scalable cloud platform with enhanced security and database support. | 3–4 Months      | Improved accessibility across different platforms. |